Algorithm
Inputs: Sensor information, textual descriptions
BEGIN:

1. Analyze the papers describing the system
2. Analyze the desired outputs of the system (e.g., through appendix section, details from paper, video demonstration etc.)
3. Analyze different risk assessment techniques (e.g., HAZOP, HARA, STPA, FMEA)
4. Select a suitable risk assessment technique
5. Analyze the selected suitable risk assessment technique
6. Perform the selected risk assessment technique to identify the hazards and safety goals associated with the system
7. Create the safety case
    a. 7.1: Specify the top goal (e.g., The system is acceptably safe)
    b. 7.2: Identify/Instantiate the relevant contexts and assumptions surrounding your system
    c. 7.3: Decompose the top goal into a strategy
    d. 7.4: Further identify and decompose into sub-goals
        i. 7.4.1: Create a strategy for each sub-goal created
    e. 7.5: Divide the sub-goal into specific parts of the system that are essential for its deployment (e.g., instructions, sensor, language capabilities) by deriving it from system functions obtained from the risk assessment technique
    f. 7.6: For each sub-goal obtained from the derivation of the system
        i. 7.6.1: Instructions
            1. 7.6.1.1: Decompose instructions into different sub-goals showcasing possible outcomes from instructions (e.g., Follow if safe, reject if dangerous)
            2. 7.6.1.2: Follow Instructions
                a. 7.6.1.2.1: Create a strategy showcasing what the system will do if instructions are to be followed
                b. 7.6.1.2.2: For each of the hazards identified from the risk assessment technique
                    i. 7.6.1.2.2.1: If further decomposition is needed to specify the strategy on how the system will follow the instructions
                        1. 7.6.1.2.2.1.1: While parent-goal is still not specific
                            a. Create a strategy for the parent-goal
                            b. Further decompose into sub-goals
                    ii. 7.6.1.2.2.2: Attach a solution to the goal
            3. 7.6.1.3: Reject instructions
                a. Follow decomposition as showcased in 7.6.2
        ii. 7.7: Sensor
            1. 7.7.1: Create a sub-goal detailing what sensors are used for the specific system
            2. 7.7.2: If sensor is not specific on how it works detailing that its specifications

                a. 7.7.2.1: Create a strategy detailing the specification
                    i. 7.7.2.1.1: While specification is not detailed on how it is able to assure that the sensor is safe to use
                        1. a. Decompose goal into further sub-goals
                        2. b. Create strategies
                    ii. 7.7.2.1.2: Attach a solution to the goal
        iii. 7.8: VLA Model capabilities
            1. 7.8.1: Create a sub-goal showcasing the specifications of the language model used for the goal
                a. 7.8.1.1: If there is further techniques used on the model (e.g., Fine-tuning, RAG-retrieval)
                    i. 7.8.1.1.1: Create a strategy showcasing what optimizations were made
                    ii. 7.8.1.1.2: Create a sub-goal detailing the technique used
                    iii. 7.8.1.1.3: While there are training adjustments or optimizations used
                        1. a. Create the necessary strategies to handle training adjustments or optimizations used
                        2. b. Create the necessary goals
                    iv. 7.8.1.1.4: Attach a solution to the goal
        iv. ….

END